

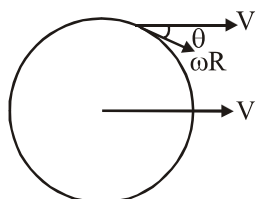
Solution :-

16. It is evident from the definition of a rigid body that no real body is truly rigid. Since real bodies deform under the influence of forces. But in many situation the deformations are negligible.
17. In pure translational motion at any instant of time all particles of the body have same velocity.
18. Movement of the axis of the fan blades (OX) around the axis known as precession axis (OY)
19. We may characterise pure rotation by all part of the body having same angular velocity at any instant of time.
21. $d_1 F_1 = d_2 F_2$ or load arm $\times$ load = effort arm $\times$ effort

$$M.A = \frac{F_1}{F_2} = \frac{d_2}{d_1}$$

22. Analogue of mass in rotational motion is moment of inertia $I = mr^2$
24. It is applicable only for lamina (2D bodies)
25. It is not necessary that the CM of a body lies at the intersection of the orthogonal axis (in theorem it is not specify)
26. This theorem is applicable to a body of any shape.
28. $\frac{K_{trans.}}{K_{Total}} = \frac{1}{1 + \frac{k^2}{R^2}} \Rightarrow \frac{1}{1 + \frac{2}{3}} \Rightarrow \frac{1}{5/3} = \frac{3}{5}$
 $= \frac{3}{5} \times 100\% \Rightarrow 60\%$
29. We apply conservation of mechanical energy of rolling body i.e., there is no loss of energy due to friction as the work done by friction force is zero.
30. As no external torque acts on planet so angular momentum will be conserved.
31. B $\rightarrow$ In pure rolling velocity of point of contact is zero.
 D $\rightarrow$ As speed of any point

$$= \sqrt{V^2 + (\omega R)^2 + 2V\omega R \cos \theta}$$



32. $L = \vec{r} \times \vec{p}$
 $= mvd (-\hat{k})$

It will remain constant.

33. As, $\tau = I\alpha$

So, Hard boiled egg will have less moment of inertia than raw egg.

34. $\vec{\tau} = \vec{r} \times \vec{F}$ so, r will increase if handle is wide hence torque will increase.

35. If axis of rotation is outside the body then every point will move around it.

36. Cat stretches its feet to increase the moment of inertia so, ω will decrease & Energy will decrease

$$R.K.E. = \frac{L^2}{2I}$$

37. As temperature will increase, radius will increase so moment of inertia will increase but angular momentum will be constant.

$$L = I\omega$$

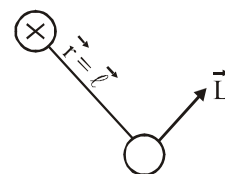
I increase, ω decreases

39. As tube is rotated the water will move with wall of the tube so moment of inertia will increase.

40. As, ice melt it will flow so, mass will disburse hence, moment of inertia will increase so, angular velocity of earth will decrease because angular momentum will be conserved so, duration of day and night will increase.

41. $\vec{L} = \vec{r} \times \vec{p}$

So, dirⁿ will change with time, but magnitude will be constant



$$|\vec{L}| = mv\ell$$

42. As, force will be $Mg \cos \theta$ perpendicular to rod so, $\tau = mg \cos \theta \ell / 2$
 It will change as θ will change

43. $\frac{R.kE}{T.kE} = \frac{\frac{1}{2}I\omega^2}{\frac{1}{2}Mv^2}$

$$\text{For, Hollow Cylinder} = \frac{\frac{1}{2}MR^2V^2/R^2}{\frac{1}{2}MV^2} = 1$$